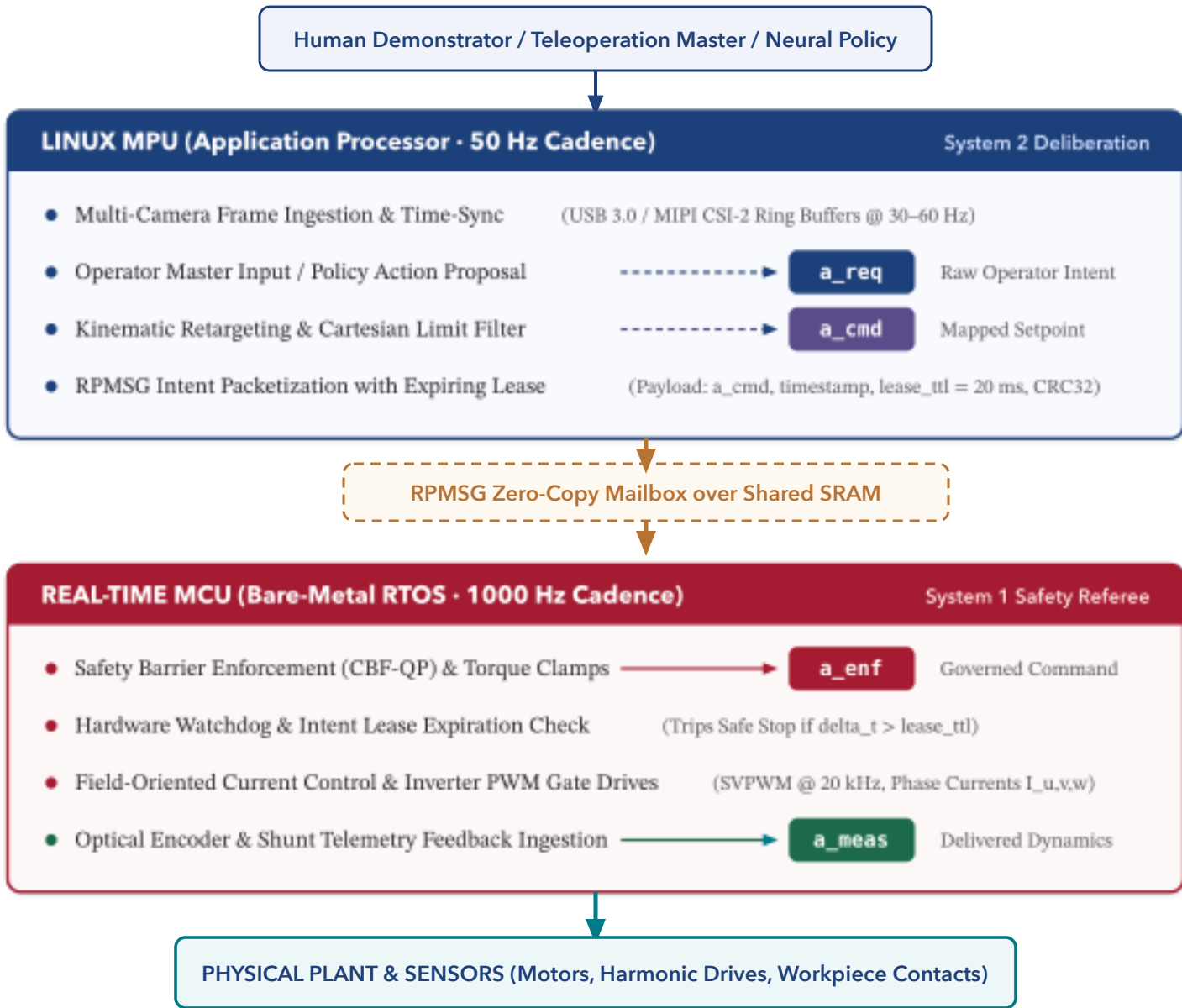




THE 4-STREAM TUPLE

a_req Requested

Raw operator / policy intent prior to software checks.

a_cmd Mapped

Cartesian/joint setpoint after kinematic retargeting.

a_enf Enforced

Safety-governed command after barrier/current clamps.

a_meas Measured

Actual delivered response (encoders, current shunts).

LOGGING INVARIANT:

Logging only a_req bakes unexecutable drift into loss.

Logging only a_meas bakes mechanical compliance and backlash as human skill.